

Workshop on Advanced Sampling Methodologies

Reporting with Quarto

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1. Install Quarto

- Download and install from <https://quarto.org/docs/get-started/>.
- Ensure R is installed if you plan to execute code chunks.

Get Started

Install Quarto, then check out the tutorials to learn the basics.

Step 1

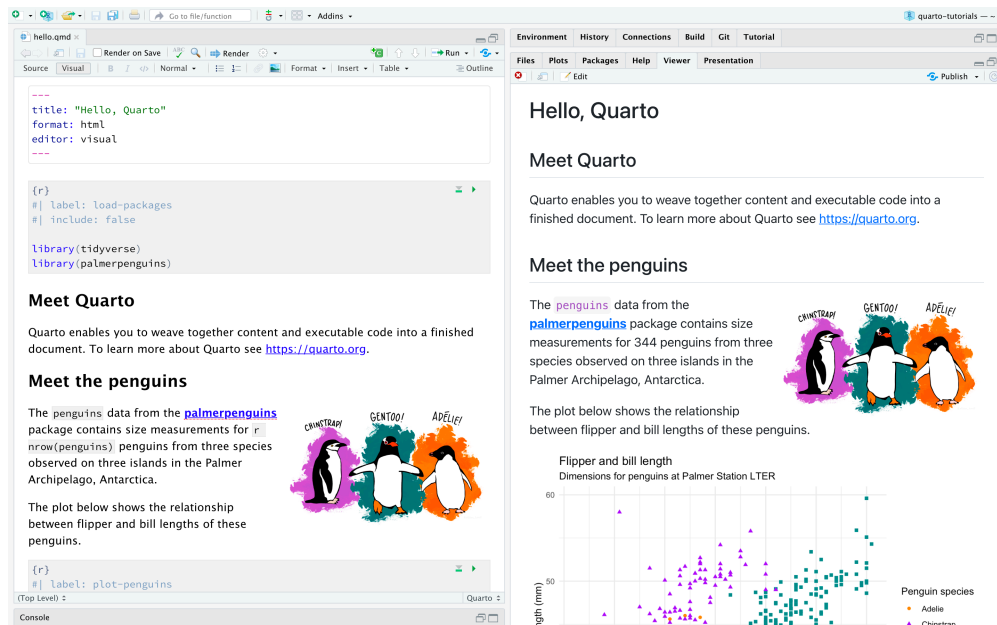
Install Quarto

Download Quarto CLI
1.8.24 (Mac OS) | Date: Sep 5, 2025

Platform	Download	Size	SHA-256
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2. Create or open a new document

- Create a new document, or
- Download and open an existing document.



 Try click on the button **Render**.

3. Two main editing modes

Quarto lets you choose between a **Visual Editor** and a **Source (Code) Editor**.

Visual Editor

- WYSIWYG interface similar to a word processor.
- Ideal for those who prefer formatting buttons and live preview.
- Features include toolbar for headings, lists, tables, and inserting code chunks.

Source (Code) Editor

- Plain-text editing of the `.qmd` file.
- Preferred by users comfortable with Markdown and YAML.
- Works well with version control systems like Git.

 Try to inspect the document. You can switch between Visual and Source modes at any time; both modes save the same underlying `.qmd` text.

4. Document structure elements

Quarto uses **YAML front matter** for metadata and **Markdown** for content.

YAML Front Matter

Example:

```
1 ---
2 title: "My Analysis"
3 author: "Jane Doe"
4 date: "2025-09-15"
5 format: html
6 ---
```

- Sets metadata such as title, author, date, and output format.

Headings

- # Level 1
- ## Level 2
- ### Level 3

Lists

- Bullet list:
 - Item one
 - Item two
- Numbered list:
 1. First
 2. Second

 Try to add headings and list in the document.

Code Chunks

- Supports many languages (R, Python, Julia, Bash, etc.).
- Chunk options (e.g., `echo`, `eval`, `fig.width`) modify output.

```
1 summary(cars)
```

speed		dist	
Min.	: 4.0	Min.	: 2.00
1st Qu.:	12.0	1st Qu.:	26.00
Median	:15.0	Median	: 36.00
Mean	:15.4	Mean	: 42.98
3rd Qu.:	19.0	3rd Qu.:	56.00
Max.	:25.0	Max.	:120.00

 Try to find code chunks in the document.

Callouts

Create callouts for tips or warnings: `::: {.callout-note}` Note that there are five types of callouts, including: **note**, **warning**, **important**, **tip**, and **caution**. `:::`



Tip with Title

This is an example of a callout with a title.



Expand To Learn About Collapse

This is an example of a 'folded' caution callout that can be expanded by the user. You can use `collapse="true"` to collapse it by default or `collapse="false"` to make a collapsible callout that is expanded by default.



Try to add callouts using the visual mode in the document.

Tables

Create simple tables using pipe syntax:

Column1	Column2
10	20

Images



 Try to add tables and images using the visual mode in the document.

3. Example Document Layout

```
1 ---
2 title: "Sales Analysis Report"
3 format: html
4 ---
5
6 # Introduction
7 Brief description...
8
9 ## Data Preparation
10
11 ::: {.cell}
12
13 :::
14
15
16 ## Results
17 - Key finding 1
18 - Key finding 2
```

5. Customize Appearance

Use a built-in theme:

```
```yaml
format:
 html:
 theme: cosmo
```
```

 Try different themes from <https://quarto.org/docs/output-formats/html-themes.html>.

6. Resources

- Documentation: <https://quarto.org/docs/>
- Tutorials and guides: <https://quarto.org/docs/get-started/>
- Community support via RStudio Community and GitHub discussions.